Ryerson University
Department of Mechanical and Industrial Engineering

MTL 200 – Materials Science Fundamentals

Midterm Test No.1

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INSTRUCTIONS

1. Duration is **50 minutes**.
2. This is a closed book test.
3. **Questions 1 and 2 are compulsory. Answer any two more questions from #3 to #5, and tick the answered questions at the top of next page.**
4. Back pages may be used if extra space is needed.
5. Please use an ink pen/ballpoint pen, and do not use a pencil (except for drawings if any).
6. Total marks: **40**

Questions	1	2	3	4	5	Total
Questions answered	√	√	✓	✓		
Maximum marks	10	10	10	10	10	40
Actual marks	8	7	10	8		33

MARKS: 10

1. Indicate the correct answer by circling TRUE or FALSE.

(1) Atomic weight is the weighted average of the atomic masses of an atom's naturally occurring isotopes.

TRUE / FALSE

(2) The smaller the principal quantum number, the higher the energy level.

TRUE / FALSE

(3) Valence electrons are the electrons in the outermost shell, which participate in the bonding between atoms.

TRUE / FALSE

(4) In the periodic table, all elements that are arrayed in a given column or group have similar valence electron structures, and chemical and physical properties.

TRUE / FALSE

(5) Bonding energy is the energy required to separate two atoms that are chemically bonded to each other.

TRUE / FALSE

(6) Ionic bonding is directional.

TRUE / FALSE

(7) There are two atoms in the face-centered cubic (FCC) unit cell.

TRUE / FALSE

(8) In the section of crystallographic directions and planes, the braces { } indicate a family of equivalent directions.

TRUE / FALSE

(9) The stacking sequence of the close-packed planes in the hexagonal close-packed (HCP) crystal structure is ABABABABAB....

TRUE / FALSE

(10) For an edge dislocation, the dislocation line and Burgers vector are perpendicular.

TRUE / FALSE

2. Circle the correct answer of the following questions.

(1) The three ages in the history of human civilizations which have been designated by the level of the materials development are

(a) Ice age, Copper age, and Iron age.
(b) Stone age, Copper age, and Gold age.
(c) Stone age, Bronze age, and Iron age.
(d) Ice age, Stone age, and Gold age.

(2) Three basic classifications of materials are

(a) nanomaterials, ceramics, and polymers.
(b) metals, nanomaterials, and polymers.
(c) metals, ceramics, and nanomaterials.
(d) metals, ceramics, and polymers.

(3) The maximum number of electrons accommodated by *d* subshell is

(a) 2.
(b) 6.
(c) 10.
(d) 14.

(4) Pauli exclusion principle specifies that

(a) each electron state can hold more than two electrons, which must have opposite spins.
(b) each electron state can hold more than two electrons, which must have the same spins.
(c) each electron state can hold no more than two electrons, which must have the same spins.
(d) each electron state can hold no more than two electrons, which must have opposite spins.

(5) Three types of primary/chemical bonds are

(a) Ionic, covalent, and Van der Waals bonding.
(b) Ionic, Van der Waals bonding, and metallic bonding.
(c) Van der Waals bonding, covalent, and metallic bonding.
(d) Ionic, covalent, and metallic bonding.

(6) Which of the following answers is correct for the coordination number in the unit cell of the face-centered cubic (FCC), body-centered cubic (BCC), and hexagonal close-packed (HCP) crystal structures, respectively?

(a) FCC $\rightarrow$ 12,	BCC $\rightarrow$ 8,	HCP $\rightarrow$ 12.
(b) FCC $\rightarrow$ 0.74,	BCC $\rightarrow$ 0.68,	HCP $\rightarrow$ 0.74.
(c) FCC $\rightarrow$ 4,	BCC $\rightarrow$ 2,	HCP $\rightarrow$ 6.
(d) FCC $\rightarrow 2\sqrt{2}$,	BCC $\rightarrow 4/\sqrt{3}$,	HCP $\rightarrow$ 1.633.

(7) What dependence of material properties is termed anisotropy?

(a) thickness.
(b) orientation.
(c) composition.
(d) atomic weight.

(8) Four main factors affecting the solubility of the substitutional solid solution are

(a) Atomic size, atomic weight, electronegativity, valences.
(b) Atomic size, crystal structure, atomic weight, valences.
(c) Atomic size, crystal structure, electronegativity, valences.
(d) Atomic weight, atomic size, crystal structure, electronegativity.

(9) Two basic types of grain boundaries are

(a) high-angle grain boundary and low-angle grain boundary.
(b) high-angle grain boundary and tilt grain boundary.
(c) tilt grain boundary and twist grain boundary.
(d) twist grain boundary and high-angle grain boundary.

(10) Two types of main diffusion mechanisms are

(a) Self-diffusion and interdiffusion.
(b) Self-diffusion and interstitial diffusion.
(c) Interdiffusion and impurity diffusion.
(d) Vacancy diffusion and interstitial diffusion.

(10)

MARKS: 10

3. The metal iridium (Ir) has an FCC crystal structure. If the diffraction angle for the (220) set of planes occurs at 69.22° (first-order reflection) when monochromatic X-radiation having a wavelength of 0.1542 nm is used, calculate **(a)** the interplanar spacing for this set of planes, and **(b)** the atomic radius for an iridium atom.

$2\theta = 69.22°$

$\theta = 34.61°$

$\lambda = (0.1542 \times 10^{-9})\,m$

$n = 1$

$n\lambda = 2 d_{hkl} \sin\theta$

$(1)(0.1542 \times 10^{-9}\ m) = (2)(d_{220}) \sin(34.61°)$

$d_{220} = \dfrac{(0.1542 \times 10^{-9}\ m)}{(2)\sin(34.61)}$

$= 1.36 \times 10^{-10}\ m$

$d_{hkl} = \dfrac{a}{\sqrt{h^2+k^2+l^2}}$

$d_{220} = \dfrac{a}{\sqrt{8}}$

$a = (1.36 \times 10^{-10}\ m)(\sqrt{8})$

$a = 3.84 \times 10^{-10}\ m$

For FCC

$a = (2)(\sqrt{2})R$

$R = \dfrac{a}{(2)(\sqrt{2})}$

$R = 1.36 \times 10^{-10}\ m$

8

MARKS: 10

4. Determine the approximate **density** and **atomic weight** of a new nickel-based superalloy LDS-1101 for turbine blade applications (see the article entitled *"A New Approach of Designing Superalloys for Low Density" – from the latest issue of JOM, January 2010, Vol.62, No.1, pp.48-54*). This alloy has a main composition of 6.0 wt% Al, 4.7 wt% Cr, 9.9 wt% Co, 7.1 wt% Mo, 3.0 wt% Re, 6.2 wt% Ta, and the remainder Ni. → 63.1 wt%

a) $$\rho_{ave} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2} + \frac{C_3}{\rho_3} + \dots}$$

$$\rho_{Ave} = \frac{100}{\frac{C_{Al}}{\rho_{Al}} + \frac{C_{Cr}}{\rho_{Cr}} + \frac{C_{Co}}{\rho_{Co}} + \frac{C_{Mo}}{\rho_{Mo}} + \frac{C_{Re}}{\rho_{Re}} + \frac{C_{Ta}}{\rho_{Ta}} + \frac{C_{Ni}}{\rho_{Ni}}}$$

$$\rho_{Ave} = \frac{100}{0.086} = 814.34 \ g/cm^3$$

$$A_{ave} = \frac{100}{\frac{C_{Al}}{A_{Al}} + \frac{C_{Cr}}{A_{Cr}} + \frac{C_{Co}}{A_{Co}} + \frac{C_{Mo}}{A_{Mo}} + \frac{C_{Re}}{A_{Re}} + \frac{C_{Ta}}{A_{Ta}} + \frac{C_{Ni}}{A_{Ni}}}$$

$$= \frac{100}{0.0168} = 59.5136 \ amu$$

MARKS: 10

5. A sheet of steel 1.5 mm thick has nitrogen atmospheres on both sides at 1200°C and is permitted to achieve a steady-state diffusion condition. The diffusion coefficient for nitrogen in steel at this temperature is 6×10^{-11} m^2/s, and the diffusion flux is found to be 1.2×10^{-7} kg/m^2-s. Also, it is known that the concentration of nitrogen in the steel at the high-pressure surface is 4 kg/m^3. How far into the sheet from this high-pressure side will the concentration be 2.0 kg/m^3? Assume a linear concentration profile.

$$J = -D \frac{\Delta C}{\Delta X}$$

$$1.20\times10^{-7}\ kg/m^2\cdot s = -(6\times10^{-11}\ m^2/s)\frac{(2.0-4.0)\ kg/m^3}{\Delta x}$$

$$\Delta x = \frac{1.2\times10^{-10}}{1.2\times10^{-7}} = (1.00\times10^{-3}\ m)$$